Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Digital Signal and Image Processing (01CT1513)	**Aim : Simulate cross correlation and autocorrelation on discrete time signals.**	
Lab Experiment : 3	**Date: 30/07/2026**	**Enrolment No:92400133050**

Github Liink : https://github.com/krishmendpara3-cyber/Digital-Signal-and-Image-Processing/blob/main/Experiment%203/EX%20-%203_050.pdf

Aim : Simulatecross correlation and autocorrelation on discrete time signals.

Code :

```
import numpy as np

import matplotlib.pyplot as plt


def cross_correlation(signal1, signal2):

    # Compute the cross-correlation

    cross_corr = np.correlate(signal1, signal2, mode='full')

    return cross_corr


def autocorrelation(signal):

    # Compute the autocorrelation

    auto_corr = np.correlate(signal, signal, mode='full')

    return auto_corr


# Define the discrete-time signals

signal1 = np.array([1, 2, 3, 4, 5])

signal2 = np.array([2, 4, 6, 8, 10])


# Compute the cross-correlation

cross_corr = cross_correlation(signal1, signal2)


# Compute the autocorrelation

auto_corr = autocorrelation(signal1)
```

```
# Plot the cross-correlation and autocorrelation signals
plt.figure(figsize=(10, 6))
plt.subplot(2, 1, 1)
plt.stem(auto_corr)
plt.title('Autocorrelation')
plt.xlabel('Time Lag')
plt.ylabel('Magnitude')

plt.subplot(2, 1, 2)
plt.stem(cross_corr)
plt.title('Cross-correlation')
plt.xlabel('Time Lag')
plt.ylabel('Magnitude')

plt.tight_layout()
plt.show()
```

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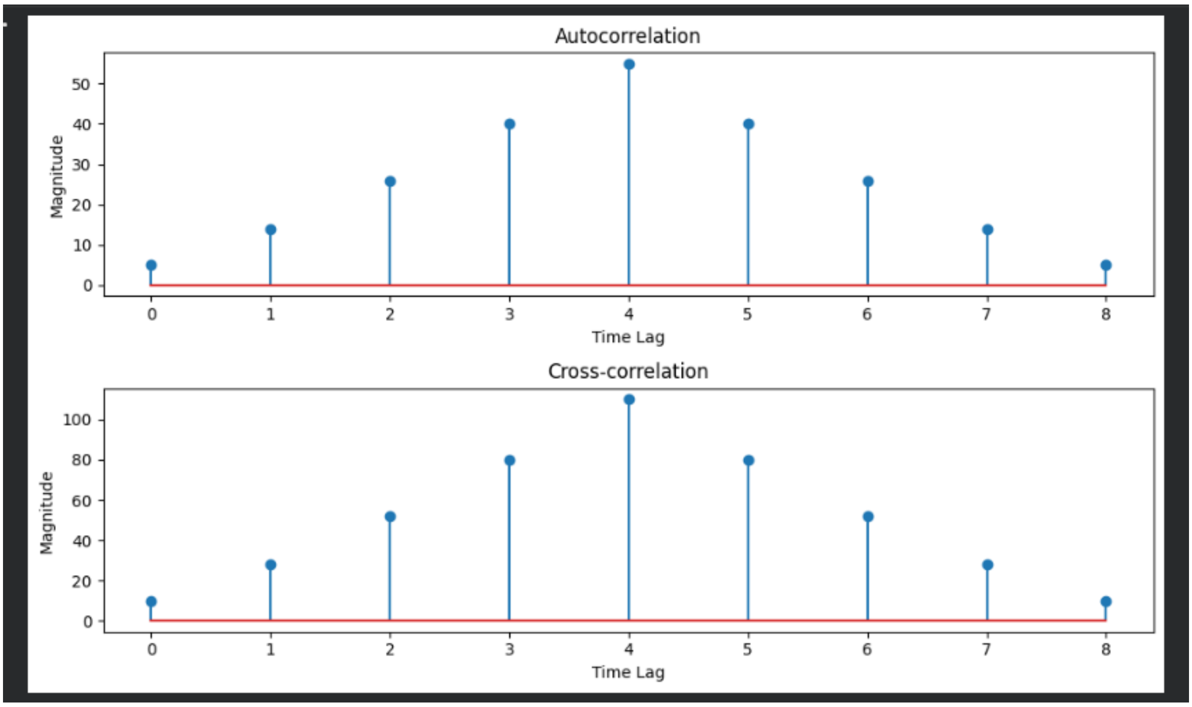
Output :



Conclusion : In this experiment We learned how to find autocorrelation and cross-correlation of discrete-time signals using Python. The autocorrelation result showed the similarity of a signal with itself at different time lags, producing a maximum value at zero lag. The cross-correlation result measured the similarity between two different signals and demonstrated their relationship across various time shifts.

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Calculation :

DATE :
PAGE :

Experiment 3

$x(n) = [1, 2, 3, 4, 5]$ $n = 0, 1, 2, 3, 4$
$y(n) = [2, 4, 6, 8, 10]$ $n = 0, 1, 2, 3, 4$

Auto Correlation

$$R_{xx}[K] = \sum_{m=-\infty}^{\infty} x[m]\,x[m-K]$$

K	Calculation	$R_{xx}(K)$
-4	1×5	5
-3	$1\times4 + 2\times5$	14
-2	$1\times3 + 2\times4 + 3\times5$	26
-1	$1\times2 + 2\times3 + 3\times4 + 4\times5$	40
0	$1^2 + 2^2 + 3^2 + 4^2 + 5^2$	55
1	$2\times1 + 3\times2 + 4\times3 + 5\times4$	40
2	$3\times1 + 4\times2 + 5\times3$	26
3	$4\times1 + 5\times2$	14
4	5×1	5

Auto Correlation : [5, 14, 26, 40, 55, 40, 26, 14, 5]

<table>
<tr><td rowspan="1">Marwadi University</td><td colspan="2">Marwadi University
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</table>

DATE :
PAGE :

Cross Correlation

$$R_{xy}[K] = \sum_{-\infty}^{\infty} x[m]\,y[m-K]$$

K	Shifted Signal $x(n-1)$	Overlapping Sum Calculation	$R_{xy}[K]$
-4	[0,0,0,0,2,4,6,8,10]	1x10	10
-3	[0,0,0,2,4,6,8,10,0]	1x8 + 2x10	28
-2	[0,0,2,4,6,8,10,0,0]	1x6 + 2x8 + 3x10	52
-1	[0,2,4,6,8,10,0,0,0]	1x4 + 2x6 + 3x8 + 4x10	80
0	[2,4,6,8,10,0,0,0,0]	1x2 + 2x4 + 3x6 + 4x8 + 5x10	110
1	[0,2,4,6,8,10,0,0,0]	2x2 + 3x4 + 4x6 + 5x8	80
2	[0,0,2,4,6,8,10,0,0]	3x2 + 4x4 + 5x6	52
3	[0,0,0,2,4,6,8,10,0]	4x2 + 5x4	28
4	[0,0,0,0,2,4,6,8,10]	5x2	10

Cross Correlation : [10, 28, 52, 80, 110, 80, 52, 28, 10]